

⚡ REAL INTERVIEW STORY

A candidate got
rejected at Amazon.

His answer was **CORRECT**.
He still got
REJECTED.

Here's the one pattern that
would have saved him.

SWIPE TO SEE WHY →

THE PROBLEM

What the interviewer actually asked:

*"Find the maximum sum of **K consecutive elements** in an array of **5,00,000 numbers.**"*

He knew the answer. He wrote the code.

He got the right output.

And then he got TLE. Time Limit Exceeded.

The interviewer wasn't testing his answer.

He was testing his

pattern recognition.



WRONG WAY

Brute Force

$O(n^2)$ – Two nested loops

```
for (int i = 0; i <= n - k; i++) {  
    long sum = 0;  
    for (int j = i; j < i+k; j++) {  
        sum += arr[j];  
        // recalculates every time  
    }  
    maxSum = Math.max(maxSum, sum);  
}
```

2847 ms

On 5,00,000 elements → TIME LIMIT EXCEEDED → REJECTED 



RIGHT WAY

Sliding Window

$O(n)$ – One loop only

```
// Step 1: First window
for (int i = 0; i < k; i++)
    windowSum += arr[i];
maxSum = windowSum;

// Step 2: Slide the window
for (int i = k; i < arr.length; i++) {
    windowSum += arr[i] - arr[i-k]; ✨
    maxSum = Math.max(maxSum, windowSum);
}
```

3 ms

Same answer • Same input • 949x faster • HIRED 



```
windowSum += arr[i] - arr[i - k]
```

```
arr[i]
```

New element entering
the window on the right

```
arr[i - k]
```

Old element leaving
the window on the left

No inner loop

No recalculation.
Just add one, remove one.

$O(n^2) \rightarrow O(n)$

One line changes your
entire complexity class.

949x

faster on real data

2847 ms $\rightarrow$ 3 ms

Where This Pattern Appears

One pattern. 15+ interview problems solved.

Maximum sum subarray of size K

Today's problem. Now you own it.

Longest substring without repeating chars

LeetCode #3. Asked at Google, Meta, Amazon.

Minimum window substring

LeetCode #76. Hard level. Sliding window core.

Max of all subarrays of size K

Amazon favourite. Uses deque + sliding window.

Pro tip: When you see "contiguous" "subarray" "window" "substring"
→ think Sliding Window first.

The interviewer was NOT
testing if you knew the
answer.

He was testing if you knew the
pattern.

Brute Force

You studied the problem.

You got the answer.

You got TLE.

Sliding Window

You recognized the pattern.

You wrote 10 lines.

You got hired.

This is the difference between a junior and a
senior developer in a FAANG interview room.

SAVE THIS. SHARE THIS.

Don't let your friend get rejected for this.

VIDEO

Full live demo with real timing proof
→ Link in first comment below

TELEGRAM

Daily DSA patterns on Telegram
→ t.me/DebugWithPurpose

FOLLOW

Follow for one pattern every day
→ 90 days = interview ready

Anupam Kumari

@DebugWithPurpose